

□ Oracle Physical Standby Setup – Short Notes for Interview (Hinglish)

□ Basic Concepts

- **Primary DB:** Main production database.
- **Standby DB:** Backup/DR copy of primary DB, live updated via Redo logs.
- **Data Guard:** Oracle ka high availability solution.
- **Standby Type:** Physical Standby.
- **Protection Mode:** Maximum Performance.
- **Tools used:** Oracle 12c, ASM, RMAN, Putty, VirtualBox.

□ High-Level Setup Steps

1□ Practice Environment Ready Karna

- 2 clones banao: ek “Primary DB” aur doosra “Standby DB”.
- Standby VM ka **hostname/IP** alag hona chahiye (e.g., srv2).
- **MAC address error fix** karo (/etc/udev/rules.d/70-persistent-net.rules).
- Standby system pe purana DB delete karo (dbca se).
- **Network settings fix** karo (system-config-network).
- Hostnames /etc/hosts aur listener.ora me update karo.

2□ Primary Database Setup

- □ Enable ArchiveLog Mode.
- □ Enable **Force Logging**.
- □ Add **Standby Redo Logs (SRLs)**.
- □ Set required parameters:
 - LOG_ARCHIVE_CONFIG, FAL_SERVER, STANDBY_FILE_MANAGEMENT, etc.
- □ Enable **Flashback**.
- □ Create Password File.
- □ Configure tnsnames.ora.

3□ Standby System Setup

- Required **directories** create karo.
- init<SID>.ora file banao (sirf DB_NAME likho).

- Primary se **password file** copy karo.
- `tnsnames.ora` aur `listener.ora` configure karo.
- Listener restart karo.

4▣ Create Standby DB (RMAN)

- Standby DB ko `STARTUP NOMOUNT` karo.
- **RMAN Duplicate command** use karo:
`DUPLICATE TARGET DATABASE FOR STANDBY FROM ACTIVE DATABASE`
- Required **SPFILE** and **control file** paths define karo.
- `nofilenamecheck` use karo (same file structure hai dono systems me).

5▣ Post Creation Tasks

- Redo Apply start karo:
`ALTER DATABASE RECOVER MANAGED STANDBY DATABASE DISCONNECT;`
- Verify redo apply and logs.
- Flashback enable karo.
- `srvctl` se standby DB ko Oracle Restart me register karo.
- Dono systems pe `ARCHIVELOG DELETION POLICY` configure karo.

▣ Interview Tips

Topic	What to Say
Why Data Guard? Physical vs Logical	For Disaster Recovery, High Availability Physical = Block-level, Logical = SQL-level
How Redo Transfer works?	Redo logs ship from Primary to Standby via LGWR/ARCH
Can we switch roles?	Yes, via Switchover and Failover
Use of FAL_SERVER?	Fetch archive logs if missing
Why Force Logging?	To make sure all transactions are logged for standby sync